QUANTIZATION OF ENERGY AND ATOMIC SPECTRA

- **Due to Planck's Contribution, energy is quantized in some systems, meaning that the system can have only certain energies and not a continuum of energies,**

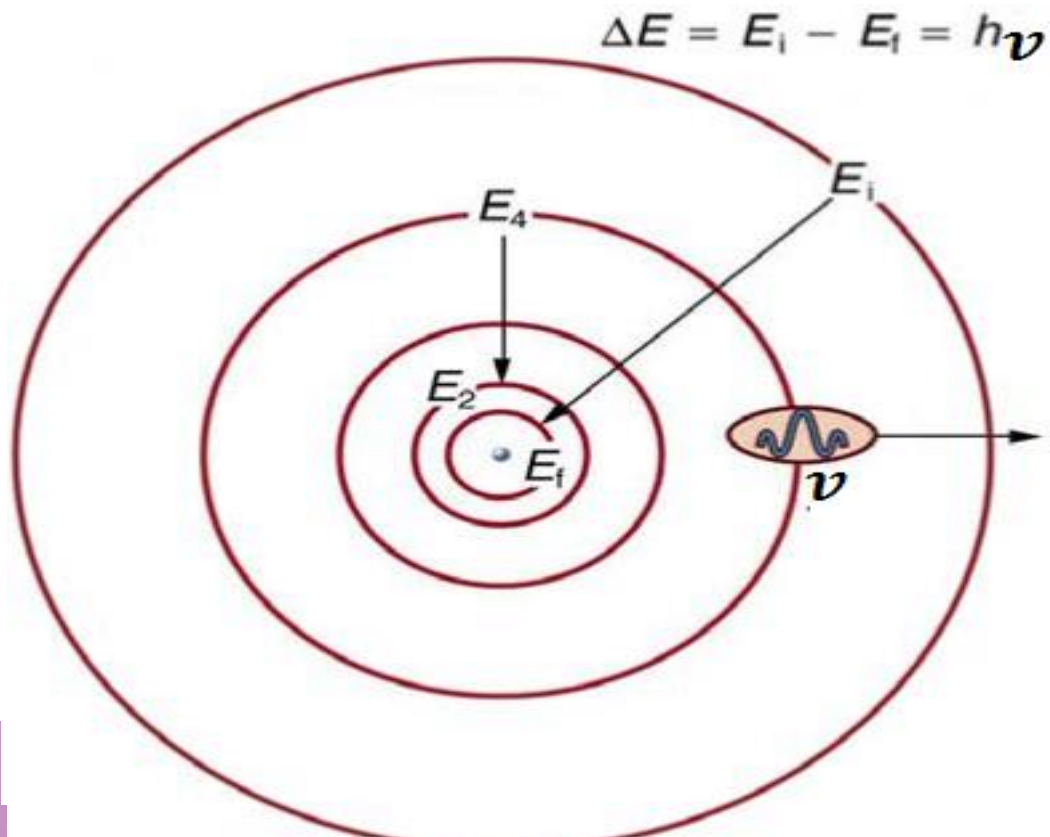
  i. e. unlike the classical case.

- **Atomic and molecular emission and absorption spectra have been known for over a century to be discrete (or quantized).**

- **Following Einstein's proposal of photons with quantized energies directly proportional to their frequencies, it became even more evident that electrons in atoms can exist only in discrete orbits.**

Bohr's Solution for Hydrogen Atom

The planetary model of the atom, as modified by Bohr, has the orbits of the electrons quantized. Only certain orbits are allowed, explaining why atomic spectra are discrete (quantized).



The energy carried away from an atom by a photon comes from the electron dropping from one allowed orbit to another and is thus *quantized*.

This is likewise true for atomic absorption of photons.

Bohr derived the formula for the hydrogen spectrum using basic physics, the planetary model of the atom, and some very important new proposals:

1. **His first proposal is that only certain orbits are allowed: we say that**

   the orbits of electrons in atoms are quantized.

   Each orbit has a different energy, and electrons can move to a higher orbit by absorbing energy and drop to a lower orbit by emitting energy.

$$\Delta E = E_i - E_f = h\nu \qquad (1)$$

Where h is Planck's constant, and ν is the frequency

If the orbits are quantized, the amount of energy absorbed or emitted is also quantized,

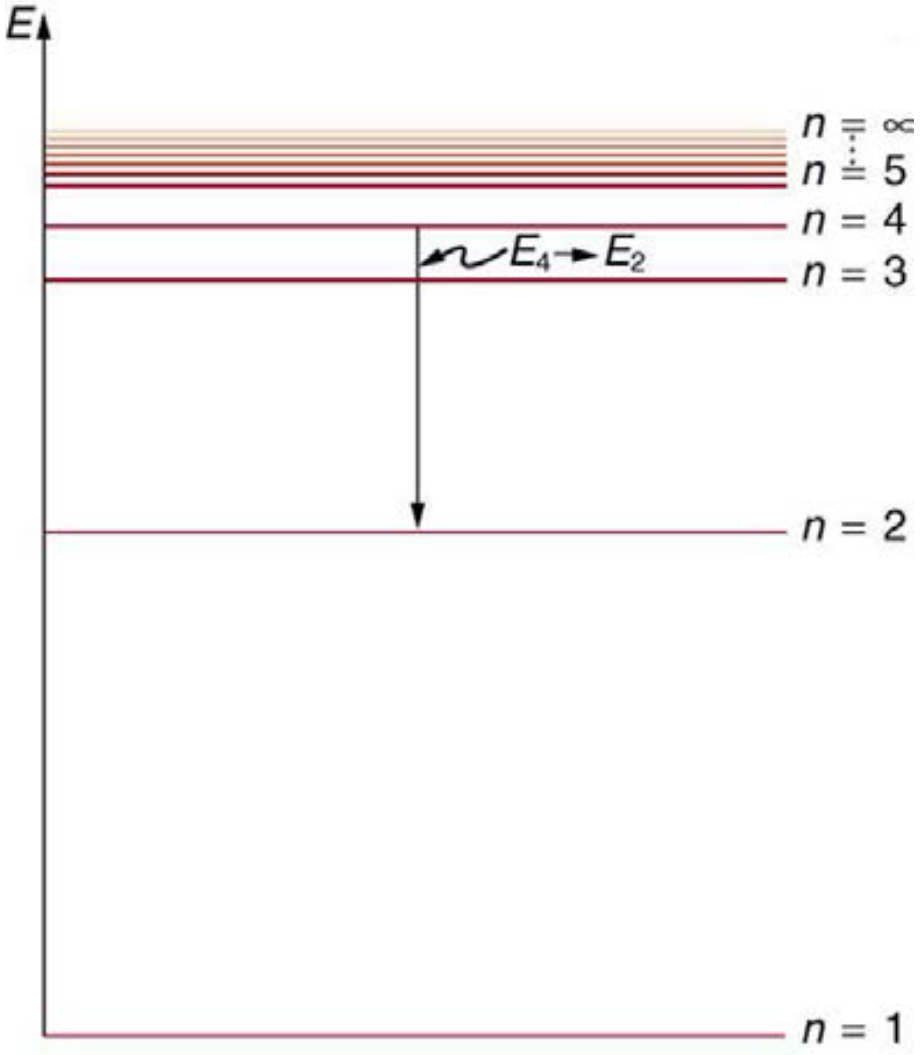
thus producing discrete spectra.

Photon absorption and emission are among the primary methods of transferring energy

into and out of atoms.

ALLOWED ENERGY-LEVEL DIAGRAM

The energy-level diagram plots energy vertically and is useful in visualizing the energy states of a system and the transitions between them.



The energy-level diagram for the hydrogen-atom electrons, showing a transition between two orbits having energies E_4 and E_2.

2. **Bohr proposed (assuming circular orbits) that the angular momentum L of an electron in its orbit is *quantized,* that is, it has only specific, discrete values.**

- **The value for *L* is given by the formula**
- $$L = m_e v\, r_n = n\frac{h}{2\pi} = n\hbar \;;(n = 1, 2, 3, \ldots), \qquad (2)$$
- **where L is the angular momentum, m_e is the electron's mass, v = tangential velocity,**
- **r_n is the radius of the nth orbit,**
- ***h* is the Planck's constant and**

$\hbar = \frac{h}{2\pi}$ ***is the reduced*** **Planck's constant.**

- **Note that angular momentum is L = Iω .**

  Where I = moment of inertia and ω= *angular velocity*
- **For a small object of mass m, at a radius r, I = mr^2 and ω = $\frac{v}{r}$, where v *= tangential velocity ;* so that**
- **L = Iω *=* $(mr^2)\frac{v}{r}$ *=* mvr *;***
- **Quantization says that this value of mvr can only be equal to $\frac{h}{2\pi}$, 2$\frac{h}{2\pi}$, 3$\frac{h}{2\pi}$, etc.**
- **using this assumption he was able to calculate the energies in the hydrogen spectrum, a novel idea, never done by anyone before!**

- **Noting that the centripetal force causing the electron to follow a circular path is supplied by the Coulomb force. To be more general, we note that this analysis is valid for any single-electron atom.**
- **So, if a nucleus has Z protons ($Z = 1$ for hydrogen, 2 for helium, etc.) and only one electron, that atom is called a hydrogen-like atom.**
- **The spectra of hydrogen-like ions are similar to hydrogen's, but shifted to higher energy by the greater attractive force between the electron and nucleus.**

The magnitude of the centripetal force is $\frac{m_e v^2}{r_n}$ **;**

while the Coulomb force is $\frac{k(Zq_e)(q_e)}{r_n^2}$ **;**

where k is a constant with a value of $8.99 \times 10^9\ N\ m^2\ C^{-2}$

- **The assumption here is that the nucleus is more massive than the electron, and the electron orbits about it. This is consistent with the planetary model of the atom.**
- **Equating the two forces:**
- $\frac{kZq_e^2}{r_n^2} = \frac{m_e v^2}{r_n}$ **(Coulomb=centripetal).** **(3)**
- **Angular momentum quantization is stated in equation (2); solving that equation for v, substituting it into the above, and rearranging the expression to obtain the radius of the orbit:**
- **From Eq. (2)**

$$v = \frac{n\hbar}{m_e r_n}$$

And substituting this into the Eq. (3) above, gives:

$$r_n = \frac{n^2}{z}\frac{h^2}{4\pi^2 m_e k q_e^2}$$

$$= \frac{n^2}{z}\frac{\hbar^2}{m_e k q_e^2}$$

- $r_n = \frac{n^2}{z} a_B$; for allowed orbits (n=1,2,3,..) (4)
- where a_B is defined to be the Bohr radius, and

- $a_B = \frac{h^2}{4\pi^2 m_e k q_e^2}$ *or* $\frac{\hbar^2}{m_e k q_e^2}$

- For the lowest orbit ($n = 1$) and for hydrogen ($Z = 1$),
- $r_1 = a_B$

Note that equation (4) shows that the orbital radius is proportional to n^2 .

Substituting equation (4) in (3):

$$a_B = \frac{h^2}{4\pi^2 m_e k q_e^2} \text{ or } \frac{\hbar^2}{m_e k q_e^2} \qquad (5)$$

$$a_B = 0.529 \times 10^{-10} \text{ m} \qquad (6)$$

- Equations (4) and (5) can be used to calculate the radii of the allowed (quantized) electron orbits in any hydrogen-like atom.
- The lowest orbit (n = 1) has the experimentally verified diameter of a hydrogen atom.

THE ELECTRON ORBITAL ENERGIES

To get the electron orbital energies, we have to note that the electron energy is the sum of its kinetic and potential energy

$$E_n = KE + PE \qquad (7)$$

Kinetic energy, $KE = (\frac{1}{2})m_e v^2$ **;** ***assuming the electron is not moving at relativistic speeds.***

Potential energy for the electron is electrical, or $PE = q_e V$ **, where** V **is the potential due to the nucleus, which looks like a point charge.**

The nucleus has a positive charge Zq_e **; thus,**

$V = \frac{kZq_e}{r_n}$, ***from the equation for the potential due to a point charge;***

where k is a constant with a value of 8.99×10^9 **N** $m^2 C^{-2}$.

But the electron's charge is negative, hence

$$PE = q_e V = -q_e \cdot \frac{kZq_e}{r_n} = -\frac{kZq_e^2}{r_n}$$

Substituting the expressions for KE and PE in (7):

$$E_n = \left(\frac{1}{2}\right) m_e v^2 - \frac{kZq_e^2}{r_n} \qquad (8)$$

substituting r_n and v from equations (4) and (3) into equation (8) for energy and simplifying:

$$E_n = -\frac{Z^2}{n^2} \frac{2\pi^2 q_e^4 m_e k^2}{h^2} \qquad (n = 1, 2, 3, \ldots) \qquad (9)$$

for the orbital energies of hydrogen-like atoms

Eq. (9) can be rewritten as:

$$E_n = -\frac{Z^2}{n^2} E_0$$

Where E_0 is the ground-state energy ($n = 1$) for hydrogen ($Z = 1$) and is given by:

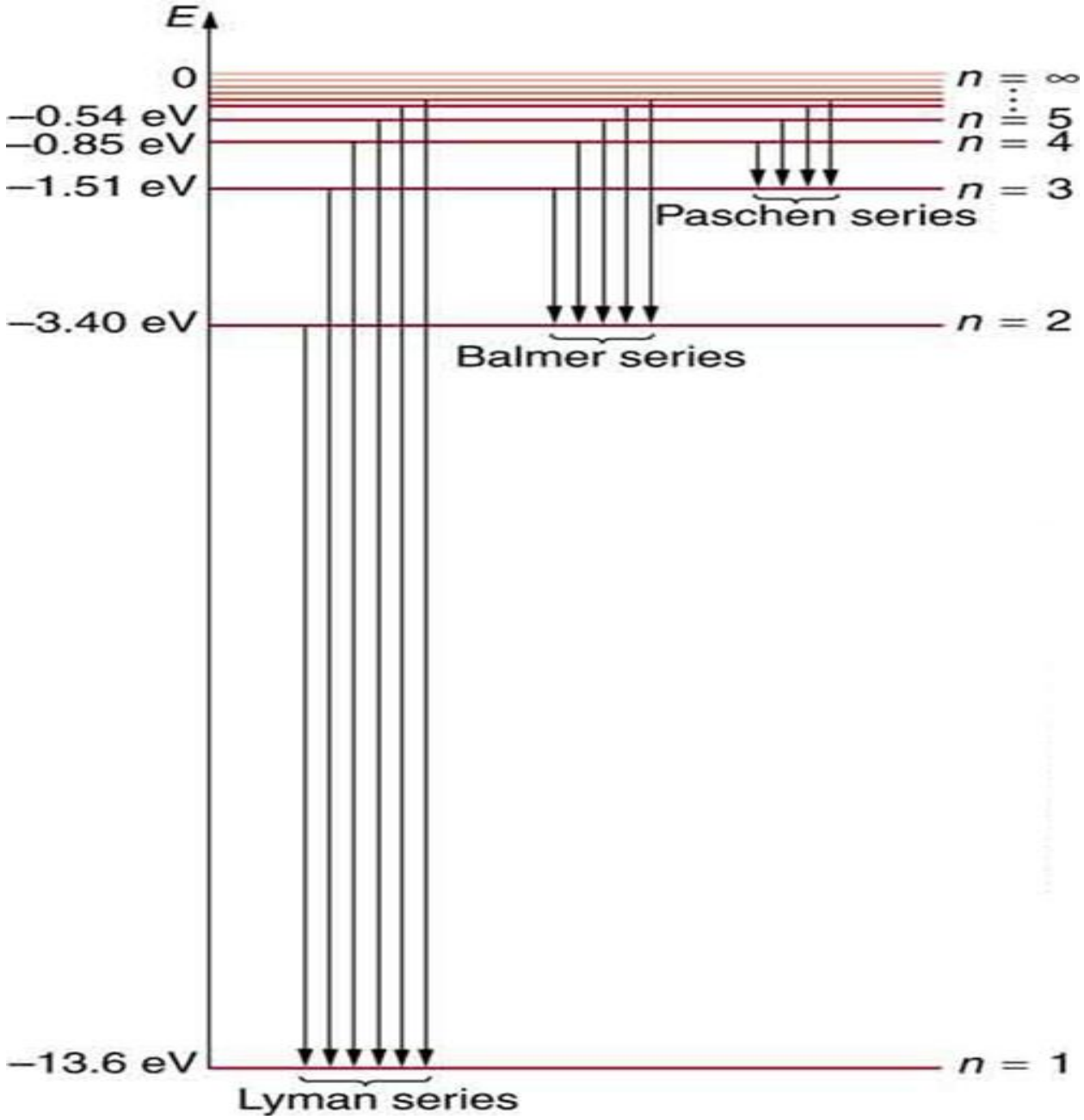
$$E_0 = \frac{2\pi^2 q_e^4 m_e k^2}{h^2} = 13.6 \text{ eV} ; \qquad (10)$$

Hence for hydrogen,

$$E_n = - \frac{13.6 \text{ eV}}{n^2} \quad (n = 1, 2, 3, \ldots) \qquad (11)$$

- **The following slide shows the energy-level diagram for hydrogen showing the**

  Lyman, Balmer, and Paschen series of transitions.
- **The orbital energies are calculated using the above equation (11), first derived by Bohr.**


E
0
−0.54 eV
−0.85 eV
−1.51 eV
−3.40 eV
−13.6 eV
n = ∞
n = 5
n = 4
n = 3
n = 2
n = 1
Paschen series
Balmer series
Lyman series

- **Electron total energies are negative, since the electron is bound to the nucleus, analogous to being in a hole without enough kinetic energy to escape. As n approaches infinity, the total energy becomes zero. This corresponds to a free electron with no kinetic energy, since r_n gets very large for large n , and the electric potential energy thus becomes zero.**
- **Thus, 13.6eV is needed to ionize hydrogen (to go from –13.6eV to 0, or unbound), an experimentally verified number.**
- **Given more energy, the electron becomes unbound with some kinetic energy.**
- **For example, giving 15.0eV to an electron in the ground state of hydrogen strips it from the atom and leaves it with 1.4eV of kinetic energy.**

Finally, we shall consider the energy of a photon emitted in a downward transition, given by the equation to be:

- $$\Delta E = h\nu = E_i - E_f \qquad (12)$$

Substituting equation (11) into (12):

$$h\nu = (13.6\ \text{eV})\left[\frac{1}{n_f^2} - \frac{1}{n_i^2}\right] \qquad (13)$$

Dividing both sides of (13) by hc gives:

$$\frac{h\nu}{hc} = \frac{\nu}{c} = \frac{1}{\lambda} = \frac{13.6\ \text{eV}}{hc}\left[\frac{1}{n_f^2} - \frac{1}{n_i^2}\right] \qquad (14)$$

It can be shown that the factor

$$\frac{13.6\ \text{eV}}{hc} = \frac{(13.6\ \text{eV})\ (1.602 \times 10^{-19}\ \text{J/eV})}{(6.626 \times 10^{-34}\ \text{J.s})\ \ (3 \times 10^{8}\ \text{m/s})}$$

$$= 1.097 \times 10^{7}\ \text{m}^{-1} = R \qquad (15)$$

Where R is the Rydberg constant.

- **Hence using Bohr's assumptions has led to the derivation of the formula first proposed by Balmer years earlier as a recipe to fit experimental data:**

$$\frac{1}{\lambda} = R\left[\frac{1}{n_f^2} - \frac{1}{n_i^2}\right] \qquad (16)$$

- **We see that Bohr's theory of the hydrogen atom answers the question as to why this previously known formula describes the hydrogen spectrum.**
- **It is because the energy levels are proportional to $\frac{1}{n^2}$, where n is a non-negative integer.**
- **A downward transition releases energy, and so n_i must be greater than n_f .**
- **The various series are those where the transitions end on a certain level.**
- **For the Lyman series, $n_f = 1$; that is, all the transitions end in the ground state.**
- **For the Balmer series, $n_f = 2$, so all the transitions end in the first excited state; and so on.**
- **Something new has emerged - *angular momentum is quantized !***

Triumphs of the Bohr Theory:

Bohr did what no one had been able to do before:

1. **He explained the spectrum of hydrogen.**
2. **He correctly calculated the size of the atom from basic physics.**
3. **Some of his ideas are broadly applicable. Electron orbital energies are quantized in all atoms and molecules.**
4. **Angular momentum is quantized.**

The electrons do not spiral into the nucleus, as expected classically (*accelerated charges radiate, so that the electron orbits classically would decay quickly, and the electrons would sit on the nucleus; matter would collapse*).

These are major triumphs.

Limits of the Bohr Theory

1. It cannot be applied to multi-electron atoms, even one as simple as a two-electron helium atom.
2. Bohr's model is what we call *semi classical*. The orbits are quantized (*non classical*) but are assumed to be simple circular paths (*classical*).
3. As quantum mechanics was developed, it became clear that there are no well-defined orbits; rather, there are clouds of probability.
4. Bohr's theory also did not explain that some spectral lines are doublets (split into two) when examined closely.

CONCLUSION:

- It should be noted that Bohr did not fail.
- Rather, he made very important steps along the path to greater knowledge and laid the foundation for all of atomic physics that has since evolved.

PHY211 ASSIGNMENT

NOTE: Always define each parameter in each of your equations, and tag your relevant equations clearly

From the expression for the total energy of the electron as the sum of is kinetic and potential energy, *assuming that the electron is not moving at relativistic speeds*;

i. Clearly show that the orbital energies of a hydrogen-like atom or ion with atomic number Z is given by

$$E_n = -\frac{Z^2}{n^2} E_0$$

Where E_0 is the ground state energy.

ii. Show that the ground state energy for hydrogen atom is

E_0 = 13.6 eV

iii. Show that the energy of a photon of frequency υ emitted in a downward transition is given by

$$h\upsilon = (13.6eV)\left[\frac{1}{n_f^2} - \frac{1}{n_i^2}\right]$$

Where subscripts *i* and *f* denote *initial* and *final* energy levels.